26F2 Homework

Phrases and Clauses and Irrational Numbers and Circles

What will homework look like?

Every week, we will give you 6–8 questions to work on (about 30 minutes), split between English language arts and mathematics. Some questions will be more challenging than others.

Try your best to submit your work before the start of the next lesson, but remember that homework is optional. You will sometimes see additional practice worksheets. You do not need to complete those; we will only grade the documents labeled "26F*X* Homework."

Homework will be graded out of eight points; a tutor will give you feedback on your work. Feel free to resubmit and correct your mistakes as many times as needed.

English Language Arts: Phrases and Clauses

1. Which choice correctly describes the bolded group of words? **[1]**
   Maya stayed after school **because she needed to finish her project.**
   - **A.** A phrase functioning as an adjective
   - **B.** A phrase functioning as an adverb
   - **C.** A dependent clause functioning as an adverb
   - **D.** An independent clause functioning as a noun

   Justify your answer here.

2. Which sentence correctly uses the introductory modifier? **[1]**
   - **A.** After studying for three hours, the test seemed much easier to Lena.
   - **B.** After studying for three hours, Lena found the test much easier.

C. After studying for three hours, the answers on Lena's test came quickly.

D. After studying for three hours, there were only a few difficult questions.

Justify your answer here.

3. What is the sentence structure of the following sentence? **[1]**
   Although the storm had ended, Marcus closed the windows and turned off the fan.
   A. Simple
   B. Compound
   C. Complex
   D. Compound-complex
   Justify your answer here.

4. What function does the bolded clause serve in the sentence? **[1]**
   The bicycle **that was leaning against the fence belongs to my neighbor.**
   A. Noun
   B. Adjective
   C. Adverb
   D. Independent clause
   Justify your answer here.

Mathematics: Irrational Numbers and Circles

Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. You may not use a calculator. For all questions in this part, a correct answer with no work shown will receive only **0.5 point**.

5. Ruth was able to run $\frac{45}{4}$ miles in $\frac{6}{5}$ hours. Benjamin was able to run $\frac{8}{3}$ that distance in $\frac{24}{5}$ hours. Write an expression for Benjamin's speed in miles per hour in terms of Ruth's speed in miles per hour, *r*. **[1.5]**
   Write your answer here.

6. Which of the following expressions represents a number that is rational but not an integer? **[1]**

A. $\left(\frac{\pi}{97}\right)^{67}$

B. $\sqrt{\frac{25}{16}}$

C. $\sqrt{125} - \sqrt[3]{64}$

D. $-\sqrt{36 + 45}$

Justify your answer here.

7. An ice cream cone can hold 471 cm^3 of ice cream. The base of the cone is a circle whose area is 78.5 cm^2. Calculate the height of the cone in centimeters. Use the formula $V_{cone} = \frac{1}{3}\pi r^2 h$ and use 3.14 for π. **[1.5]**

Write your answer here.